Vishal Panuganti

MASTERS IN DATA SCIENCE | AI / ML ENGINEER

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with extensive experience developing end-to-end machine learning, computer vision, and data engineering solutions. Specialized in transitioning solutions from raw dataset preparation and model training to robust cloud deployment and production optimization. Proficient across Python, PyTorch, NVIDIA TAO Toolkit, YOLO, OCR, AWS, Azure, and modern MLOps frameworks, with a proven record of engineering scalable, business-focused AI architectures.

TECHNICAL SKILLS

Languages & Frameworks: Python, R, SQL, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, SciPy, OpenCV, NVIDIA TAO Toolkit, Ultralytics YOLO

AI/ML & Data Science: Machine Learning, Deep Learning, Generative AI, LLMs, RAG, Prompt Engineering, AI Agents, NLP, Computer Vision, OCR, Recommendation Systems, Time Series Forecasting

Cloud, DevOps & BI Tools: AWS (RDS, S3, SCT), Azure & Azure DevOps, Databricks, Docker, CI/CD, Linux, Git, Power BI, Tableau, Apache Superset, Grafana, Microsoft Fabric

PROFESSIONAL EXPERIENCE

Software Engineer — Techwave

2025-03 – Present

Hyderabad, TG

Fulton Hogan — Infrastructure Defect Detection

- Collected, prepared, and validated extensive image datasets, conducted detailed annotation quality assurance, and trained specialized computer vision models using NVIDIA TAO Toolkit, PyTorch, and Ultralytics YOLO for automated infrastructure inspection.
- Improved model performance through systematic hyperparameter tuning, confidence threshold optimization, data augmentation, and iterative evaluation, significantly reducing false detections while elevating prediction reliability.
- Designed and maintained production-ready ML pipelines encompassing image preprocessing, high-throughput inference, post-processing, automated validation, debugging, and deployment support.
- Enforced engineering best practices and version control using Git and Azure DevOps while mentoring junior engineers and collaborating with cross-functional technical teams to resolve complex machine learning bottlenecks.

ACT — Defect Monitoring & Operational BI

- Developed interactive Power BI dashboards and KPI reports utilizing DAX, Power Query, and analytical data modeling to deliver strategic operational insights for infrastructure defect monitoring.
- Architected comprehensive defect reporting solutions across Power BI, Apache Superset, and Grafana, providing real-time visibility into defect trends, project health metrics, and overall business performance.
- Optimized enterprise reporting workflows by streamlining raw data transformation, automating recurring report generation, and partnering with executive stakeholders to deploy scalable business intelligence solutions.

Data Engineer Intern — Schneider Electric

2021-06 – 2021-10

Nashville, USA

- Planned and executed the cloud migration of an on-premises Oracle Database to Amazon RDS, analyzing critical database dependencies, verifying migration readiness, and coordinating execution activities to guarantee minimal operational disruption and total data integrity.
- Performed comprehensive database schema assessments and conversions using the AWS Schema Conversion Tool (AWS SCT), converting database schemas, tables, views, indexes, and SQL objects while remediating compatibility edge cases.

- Engineered and optimized large-scale ETL workflows in Databricks by refactoring daily and monthly batch data processing scripts to ingest, transform, and update data stored in Amazon S3, enhancing pipeline maintainability and data reliability.

KEY PROJECTS

Hybrid Recommendation System

Python | Scikit-learn | Taipy

Technologies: Python, Pandas, NumPy, Scikit-learn, Cosine Similarity, Matrix Factorization (SVD), Taipy

- Architected a robust hybrid recommendation engine integrating Collaborative Filtering and Content-Based Filtering techniques to deliver highly accurate, personalized product recommendations from enterprise retail data.
- Engineered end-to-end data processing pipelines, incorporating advanced feature extraction, similarity matrix calculations, and model-based ranking algorithms.
- Developed and deployed an interactive, user-centric web interface using Taipy, enabling real-time product exploration and dynamic recommendation generation for end users.

Stock Market Forecasting System

Python | TimesFM | Plotly

Technologies: Python, Google TimesFM Foundation Model, Time Series Analysis, Jugaad Data API, Plotly, Git

- Built an automated stock market forecasting system utilizing Google's TimesFM (Time Series Foundation Model) to predict short-term and long-term asset price trajectories from historical market data.
- Designed complete data ingestion pipelines interfacing with the Jugaad Data API, implementing feature engineering, zero-shot foundation model inference, and statistical accuracy evaluation.
- Integrated interactive data visualization dashboards using Plotly to provide intuitive, data-driven financial insights for algorithmic market analysis.

EDUCATION

Masters in Data Science **2019 – 2021**

GPA: 3.4 / 4.0

B.Tech in Computer Science — Sreyas Institute of Engineering and Technology **2015 – 2019**

GPA: 7.4 / 10

CERTIFICATIONS & LANGUAGES

- **Certifications:** Microsoft Certified: Power BI Data Analyst Associate
- **Languages:** English (Fluent), Hindi (Fluent), Telugu (Fluent), Japanese (Fluent)